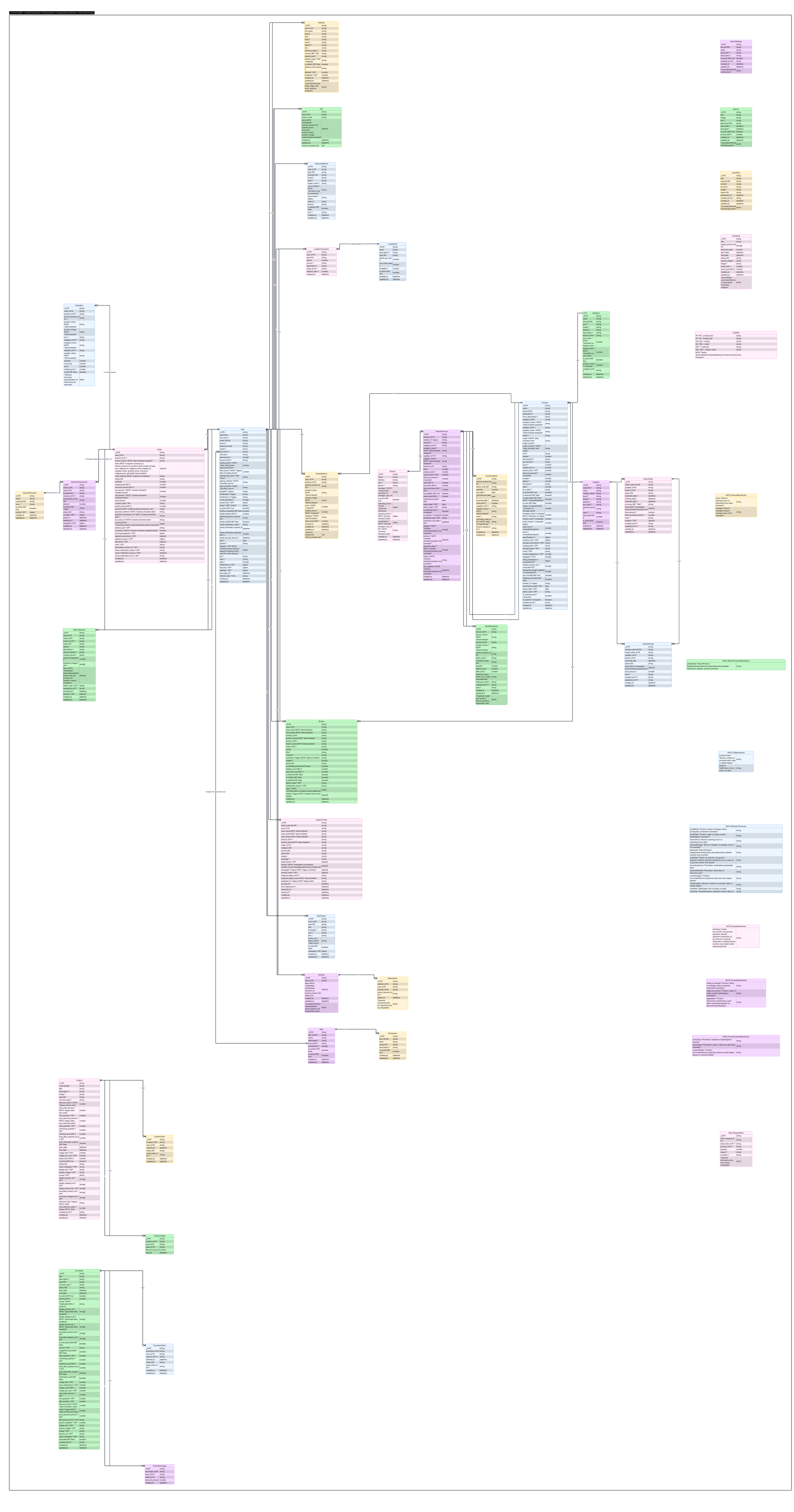

E-commerce DB ERD — Core/Auth/Catalog/Orders + Promotions/Payments + CX/Support/Reviews/Notifications + Admin Inventory & Supply

"1-N"
"rule-applied (optional)"
"1-N (if extracted)"
"1-N (project-dependent; sometimes 1-1)"
"parent-child"
"1-N (optional)"
"assigned via role_id/role_key"
"role.permissions[] includes Permission.key"

Address	
_id PK	string
user_id FK	string
full_name	string
phone	string
line1	string
line2 ?	string
ward ?	string
district ?	string
city	string
province_state ?	string
country DEF "VN"	string
postal_code ?	string
address_type ? DEF "shipping"	string
is_default DEF false	boolean
delivery_instructions ?	string
latitude ? OPT	number
longitude ? OPT	number
created_at	datetime
updated_at	datetime
"uncertain/inferred fields; aligns with order_address snapshot"	NOTE

Cart	
_id PK	string
user_id FK	string
branch_id FK	string
items NOTE "embedded: {branch_product_id*, quantity, price, unit_price, product_name, product_image} snapshot/denormalized"	object[]
created_at	datetime
updated_at	datetime
"(user_id, branch_id)"	UQ

PaymentMethod	
_id PK	string
user_id FK	string
type IDX	string
provider IDX	string
brand ?	string
last4 ?	string
holder_name ?	string
card_number ? NOTE "sensitive; may be tokenized"	string
card_holder ? legacy	string
expiry ?	string
phone ?	string
is_default DEF false	boolean
icon ?	string
created_at	datetime
updated_at	datetime

LoyaltyTransaction	
_id PK	string
user_id FK	string
type IDX	string
points	number
source ?	string
description ?	string
order_id FK ?	string
balance_after ?	number
created_at	datetime

LoyaltyRule	
_id PK	string
name	string
description ?	string
type IDX	string
points_per_unit ?	number
min_order_value ?	number
multiplier ?	number
is_active DEF true	boolean
created_at	datetime
updated_at	datetime

AdminSettings	
_id PK	string
key UQ IDX	string
value	string
group IDX ?	string
description ?	string
is_active DEF true	boolean
updated_by FK ?	string
created_at	datetime
updated_at	datetime
"uncertain/inferred config store"	NOTE

Banner	
_id PK	string
title	string
image	string
link ?	string
placement IDX	string
start_date ?	datetime
end_date ?	datetime
is_active DEF true	boolean
priority DEF 0	number
created_at	datetime
updated_at	datetime
"uncertain/inferred marketing asset"	NOTE

EventPost	
_id PK	string
title	string
slug UQ IDX	string
content	string
excerpt ?	string
image ?	string
status IDX	string
published_at ?	datetime
created_by FK ?	string
created_at	datetime
updated_at	datetime
"uncertain/inferred marketing/content"	NOTE

FlashDeal	
_id PK	string
title	string
target_product_ids FK	string[]
discount_value	number
start_date	datetime
end_date	datetime
status IDX	string
banner_image ?	string
image ?	string
stock_limit ?	number
sold_count DEF 0	number
created_at	datetime
updated_at	datetime
"aka HotDeal; uncertain/inferred or specialized Promotion subtype"	NOTE

OrderItem	
_id PK	string
order_id FK	string
product_id FK ?	string
branch_product_id FK ?	string
product_name NOTE "denormalized"	string
product_image NOTE "denormalized"	string
sku ?	string
category_id FK ?	string
category_name NOTE "denormalized"	string
supplier_id FK ?	string
supplier_name NOTE "denormalized"	string
quantity	number
unit_price	number
price	number
original_price ?	number
is_gift DEF false	boolean
"Optional extracted representation of Order.items for reporting"	NOTE

Category	
_id PK	string
name	string
slug UQ IDX	string
icon ?	string
image ?	string
banner ?	string
description ?	string
parent_id FK ?	string
sort_order ? NOTE "mismatch vs display_order"	number
display_order ? NOTE "mismatch vs sort_order"	number
is_active DEF true	boolean
product_count ? computed	number
created_by FK ?	string
created_at	datetime
updated_at	datetime

LEGEND	
PK "PK = primary key"	string
FK "FK = foreign key"	string
UQ "UQ = unique"	string
IDX "IDX = index"	string
OPT "?: optional"	string
DEF "DEF = default value"	string
NOTE "Notes: denormalized/computed/legacy/uncertain/schema mismatch"	string

Product	
_id PK	string
name	string
slug UQ IDX	string
description ?	string
short_description ?	string
category_id FK	string
category_name ? NOTE "denormalized snapshot"	string
supplier_id FK ?	string
supplier_name ? NOTE "denormalized snapshot"	string
brand ?	string
origin ? NOTE "alias mismatch with origin_country"	string
origin_country ? NOTE "alias mismatch with origin"	string
unit ?	string
weight ?	number
barcode IDX ?	string
sku UQ IDX ?	string
price ?	number
original_price ?	number
import_price ? OPT	number
discount_percent ? computed	number
images ?	string[]
gallery ?	string[]
thumbnail ?	string
tags ?	string[]
eco_label ?	string
is_active DEF true	boolean
is_featured DEF false	boolean
is_best_seller DEF false NOTE "computed/display"	boolean
is_new DEF false	boolean
rating ? computed NOTE "mismatch vs average_rating"	number
average_rating ? computed NOTE "mismatch vs rating"	number
review_count ? computed	number
total_reviews ? computed legacy	number
sold_count ? computed/snapshot	number
stock ? computed/snapshot	number
specifications ?	object
nutrition_info ? OPT	object
storage_instructions ? OPT	string
usage_guide ? OPT	string
storage_guide ? OPT	string
notes ? OPT	string
recipe_suggestions ? OPT	string[]
highlights ? OPT	string[]
rating_breakdown ? computed OPT	object
related_product_ids ? computed OPT	string[]
frequently_bought_together ? computed OPT	string[]
vat_included DEF true	boolean
shipping_excluded DEF false	boolean
master_id ? legacy	string
manufacture_date ? OPT	date
expiry_date ? OPT	date
batch_code ? OPT	string
is_expiring_soon ? computed	boolean
is_expired ? computed	boolean
created_by FK ?	string
created_at	datetime
updated_at	datetime

BranchProduct	
_id PK	string
product_id FK	string
master_id ? legacy	string
sku IDX ?	string
category_id FK ?	string
category_name ? NOTE "denormalized snapshot"	string
supplier_id FK ?	string
supplier_name ? NOTE "denormalized snapshot"	string
branch_id FK	string
price	number
original_price ?	number
import_price ? OPT	number
discount_percent ? computed	number
stock DEF 0	number
min_stock DEF 0	number
max_purchase_limit ? OPT	number
is_available DEF true	boolean
sold_count DEF 0	number
manufacture_date ? OPT	date
expiry_date ? OPT	date
batch_code ? OPT	string
is_expiring_soon ? computed	boolean
is_expired ? computed	boolean
promotion_tag ? OPT	string
promotion_end_date ? OPT	datetime
badges ? NOTE "schema mismatch/sample-only uncertain"	string[]
policies ? NOTE "schema mismatch/sample-only uncertain"	object[]
lead_time_days ? NOTE "schema mismatch/sample-only uncertain"	number
status ? NOTE "schema mismatch/sample-only uncertain"	string
last_updated ? NOTE "schema mismatch/sample-only uncertain"	datetime
created_at	datetime
updated_at	datetime

InventoryBatch	
_id PK	string
branch_product_id FK	string
batch_code IDX	string
quantity	number
exp_date ?	date
manufacture_date ?	date
received_date ?	datetime
cost_price ?	number
supplier_id FK ?	string
supplier_name ? NOTE "denormalized"	string
note ?	string
purchase_order_id FK ? NOTE "alias of ImportOrder"	string
import_receipt_id FK ?	string
created_at	datetime
updated_at	datetime

Supplier	
_id PK	string
code IDX ?	string
name	string
contact_name ?	string
email ?	string
phone ?	string
address ?	string
tax_code ?	string
payment_terms ?	string
note ?	string
total_debt DEF 0	number
is_active DEF true	boolean
created_at	datetime
updated_at	datetime

ImportOrder	
_id PK	string
order_code UQ IDX	string
supplier_id FK	string
branch_id FK	string
status IDX	string
expected_date ?	datetime
ordered_date ?	datetime
received_date ?	datetime
currency DEF "VND"	string
items NOTE "embedded: product/batch/quantity/cost"	object[]
total_amount ?	number
total_received_amount ?	number
note ?	string
timeline ? OPT	object[]
created_by FK ?	string
updated_by FK ?	string
created_at	datetime
updated_at	datetime

NOTE_BranchMismatches	
hours "Branch: opening_hours vs operating_hours alias mismatch"	string
manager "Branch: manager string vs manager_user_id FK mismatch"	string

User	
_id PK	string
username	string
full_name ?	string
email UQ IDX	string
phone ?	string
password_hash	string
avatar ?	string
role_id FK ?	string
role_key ?	string
permissions ?	string[]
branch_id FK ?	string
loyalty_points ? NOTE "alias: lotte_points legacy/mismatch"	number
lotte_points ? NOTE "legacy alias of loyalty_points"	number
membership_level ? DEF "basic"	string
signup_method ? NOTE "provider source"	string
login_provider ? NOTE "provider source"	string
googleId ? legacy	string
facebookId ? legacy	string
facebook_id ? legacy	string
social_providers ?	string[]
social_links ? OPT	object
status ? DEF "active"	string
is_active DEF true	boolean
profile_completed DEF false	boolean
wallet_balance DEF 0	number
default_payment_method FK ?	string
email_verified DEF false	boolean
email_verification_code ?	string
email_verification_expires_at ?	datetime
email_verification_attempts DEF 0	number
email_otp_last_sent_at ?	datetime
dob ?	date
gender ?	string
address ? OPT NOTE "profile primary address; separate Address entity used for multi-address"	object
bio ?	string
note ?	string
tags ?	string[]
preferences ? OPT	object
security ? OPT	object
settings ? OPT	object
last_login_at ?	datetime
refresh_token ? IDX	string
created_at	datetime
updated_at	datetime

Order	
_id PK	string
user_id FK	string
branch_id FK ?	string
branch_name ? NOTE "denormalized snapshot"	string
items NOTE "snapshot: {product_id, branch_product_id, product_name, product_image, sku, category_id, category_name, supplier_id, supplier_name, quantity, price, unit_price, original_price, gift_flag?} denormalized"	object[]
order_address NOTE "snapshot of Address"	object
status IDX	string
subtotal	number
shipping_fee DEF 0	number
discount_amount DEF 0	number
tax_amount ? NOTE "schema mismatch sample/runtime"	number
vat_percent ? NOTE "schema mismatch sample/runtime"	number
total_amount	number
coupon_code ? IDX	string
points_earned DEF 0	number
payment NOTE "{method,status,transaction_id}"	object
payment_method ? NOTE "schema mismatch alias"	string
payment_transaction_id ? NOTE "schema mismatch alias"	string
qr_code_url ? NOTE "schema mismatch alias"	string
tracking NOTE "{tracking_number,carrier,estimated_delivery,history[]}"	object
delivery_slot ? OPT	string
is_pickup ? NOTE "schema mismatch sample/runtime"	boolean
pricing_breakdown ? OPT	object
applied_promotions ? OPT	object[]
applied_coupon ? OPT	object
gift_items ? OPT	object[]
note ? OPT	string
generated_invoice_url ? OPT	string
email_notification_status ? OPT	string
email_notification_sent_at ? OPT	datetime
email_notification_error ? OPT	string
created_at	datetime
updated_at	datetime

PaymentTransaction	
_id PK	string
order_id FK	string
user_id FK	string
provider IDX	string
method_id FK ?	string
transaction_id UQ IDX	string
amount	number
currency DEF "VND"	string
status IDX	string
qr_data ? OPT	object
paid_at ?	datetime
expired_at ?	datetime
metadata ? OPT	object
created_at	datetime
updated_at	datetime

PaymentProvider	
_id PK	string
name	string
code UQ IDX	string
icon ?	string
is_active DEF true	boolean
config ? OPT	object
created_at	datetime
updated_at	datetime

ViewedHistory	
_id PK	string
user_id FK	string
product_id FK	string
branch_product_id FK ?	string
product_name NOTE "denormalized"	string
product_image NOTE "denormalized"	string
price ? NOTE "snapshot"	number
original_price ? NOTE "snapshot"	number
category ? NOTE "denormalized"	string
view_count DEF 1	number
viewed_at	datetime
created_at	datetime
updated_at	datetime
"(user_id, product_id, branch_product_id)"	UQ

Branch	
_id PK	string
name	string
address	string
city ?	string
phone ?	string
manager ? NOTE "string name; mismatch vs manager_user_id"	string
is_active DEF true	boolean
operating_hours ? NOTE "preferred canonical"	object
opening_hours ? NOTE "schema mismatch alias"	object
coordinates ? NOTE "{lat,lng}"	object
manager_user_id FK ? NOTE "schema mismatch alias"	string
created_at	datetime
updated_at	datetime

StockMovement	
_id PK	string
branch_id FK	string
branch_name ? NOTE "denormalized"	string
product_id FK	string
product_name ? NOTE "denormalized"	string
branch_product_id FK	string
batch_code ?	string
movement_type IDX	string
quantity	number
before_stock	number
after_stock	number
reference_type ? NOTE "e.g., Order, ImportOrder, ImportReceipt"	string
reference_id FK ?	string
created_by FK ?	string
note ?	string
created_at	datetime
updated_at	datetime
"Duplicate model also exists in Misc.js (schema duplication risk)"	NOTE

ImportReceipt	
_id PK	string
receipt_code UQ IDX	string
import_order_id FK	string
supplier_id FK	string
branch_id FK	string
received_date	datetime
status IDX	string
items NOTE "embedded: product/batch/quantity/cost"	object[]
total_amount ?	number
note ?	string
created_by FK ?	string
updated_by FK ?	string
created_at	datetime
updated_at	datetime

NOTE_BranchProductMismatches	
sampleOnly "BranchProduct: badges/policies/lead_time_days/status/last_updated observed in sample; schema uncertain"	string

ReturnRequest	
_id PK	string
user_id FK	string
order_id FK	string
branch_id FK ?	string
status IDX	string
reason	string
description ?	string
refund_method ?	string
contact_phone ?	string
amount_requested ?	number
evidence_images ? OPT	string[]
items NOTE "embedded ReturnRequestItem: {order_item_id?, product_id?, quantity, reason, condition}"	object[]
admin_note ? OPT	string
resolved_by FK ?	string
resolved_at ?	datetime
timeline ? OPT	object[]
created_at	datetime
updated_at	datetime

Review	
_id PK	string
user_id FK	string
user_name NOTE "denormalized"	string
user_avatar NOTE "denormalized"	string
product_id FK	string
product_name NOTE "denormalized"	string
branch_id FK ?	string
branch_name NOTE "denormalized"	string
order_id FK ?	string
rating	number
title ?	string
content ?	string
comment ? legacy NOTE "alias of content"	string
images ?	string[]
status IDX	string
is_verified_purchase DEF false	boolean
helpful_count DEF 0	number
reported_count DEF 0	number
is_featured DEF false	boolean
is_hidden DEF false	boolean
is_deleted DEF false	boolean
admin_notes ? OPT	string
moderation_reason ? OPT	string
reply ? NOTE "{content,admin_id,admin_name,replied_at}"	object
replies ? legacy NOTE "sample shows array replies"	object[]
created_at	datetime
updated_at	datetime

NOTE_CXMismatches	
reviewContent "Review: content vs comment alias; reply vs replies legacy"	string
notifLink "Notification: link vs action_url alias"	string

SupportTicket	
_id PK	string
ticket_code UQ IDX	string
user_id FK	string
user_name NOTE "denormalized"	string
user_email NOTE "denormalized"	string
user_avatar NOTE "denormalized"	string
branch_id FK ?	string
branch_name NOTE "denormalized"	string
order_id FK ?	string
category IDX	string
priority IDX	string
status IDX	string
subject	string
message ?	string
attachments ? OPT	object[]
thread ? NOTE "embedded conversation: {sender_id,role,message,attachments,created_at}"	object[]
messages ? legacy NOTE "legacy container"	object[]
internal_notes ? OPT	object[]
assigned_agent_id FK ?	string
assigned_agent_name NOTE "denormalized"	string
assigned_to ? legacy NOTE "legacy alias"	string
sla_due_at ?	datetime
first_response_at ?	datetime
resolved_at ?	datetime
closed_at ?	datetime
created_at	datetime
updated_at	datetime

NOTE_MismatchSummary	
prodRating "Product: rating vs average_rating (computed; schema/FE mismatch)"	string
prodOrigin "Product: origin vs origin_country (alias/legacy mismatch)"	string
branchHours "Branch: opening_hours vs operating_hours alias"	string
branchManager "Branch: manager vs manager_user_id FK mismatch"	string
bpSample "BranchProduct: badges/policies/lead_time_days/status/last_updated sample-only uncertain"	string
orderAlias "Order: tax_amount, vat_percent, payment_method, payment_transaction_id, qr_code_url, is_pickup runtime-only aliases"	string
promoDupScope "Promotion: duplicated scope/target defs"	string
promoValueAlias "Promotion: value alias of discount_value"	string
couponLegacy "Coupon: min_order/discount_type/max_discount_value legacy aliases"	string
reviewLegacy "Review: content vs comment; reply vs replies legacy"	string
notifAlias "Notification: link vs action_url alias"	string
stockDup "StockMovement: duplicate model in Misc.js"	string

Notification	
_id PK	string
user_id FK	string
type IDX	string
title	string
message ?	string
icon ?	string
link ?	string
action_url ? legacy NOTE "alias of link"	string
is_read DEF false	boolean
metadata ? OPT	object
created_at	datetime
updated_at	datetime

NOTE_OrderMismatches	
taxPickup "Order: tax_amount, vat_percent, payment_method, payment_transaction_id, qr_code_url, is_pickup observed in sample/runtime; schema may embed under payment/pricing"	string

Wishlist	
_id PK	string
user_id FK	string
items NOTE "embedded WishlistItem: {product_id, branch_product_id?, added_at}"	object[]
created_at	datetime
updated_at	datetime
"uncertain/inferred representation; some projects use WishlistItem table"	NOTE

WishlistItem	
_id PK	string
wishlist_id FK	string
user_id FK	string
product_id FK	string
branch_product_id FK ?	string
added_at	datetime
"Optional extracted entity for reporting; may be embedded"	NOTE

NOTE_ProductMismatches	
rating_vs_average "Product: rating vs average_rating (computed; schema/FE mismatch)"	string
origin_vs_country "Product: origin vs origin_country (alias/legacy mismatch)"	string
aggregates "Product: stock/sold_count/review_count often computed/snapshot via BranchProduct/Reviews"	string

Role	
_id PK	string
key UQ IDX	string
name	string
description ?	string
role_id IDX ?	string
permissions ?	string[]
is_system DEF false	boolean
is_active DEF true	boolean
created_at	datetime
updated_at	datetime

Permission	
_id PK	string
key UQ IDX	string
label	string
group IDX	string
description ?	string
is_active DEF true	boolean
created_at	datetime
updated_at	datetime

NOTE_PromoCouponMismatches	
promoDup "Promotion: duplicate scope/targets in schema"	string
promoValue "Promotion: value vs discount_value alias mismatch"	string
couponAliases "Coupon: min_order/discount_type/max_discount_value legacy aliases vs canonical fields"	string

Coupon	
_id PK	string
code UQ IDX	string
title	string
description ?	string
image ?	string
type IDX	string
voucher_type ?	string
discount_value ? NOTE "legacy aliases exist"	number
min_order_amount ? NOTE "legacy alias: min_order"	number
min_quantity ? OPT	number
max_discount_amount ? NOTE "legacy alias: max_discount_value"	number
total_quantity ? OPT	number
remaining_quantity ? OPT	number
claimed_count DEF 0	number
hide_after_expired_hours ? OPT	number
auto_hide_after_expired DEF false	boolean
start_date	datetime
end_date	datetime
usage_limit ? OPT	number
usage_per_user ? OPT	number
used_count DEF 0	number
is_active DEF true	boolean
status IDX	string
claim_campaign ? OPT	string
badge_text ? OPT	string
banner_image ? OPT	string
scope ? OPT	string
target_product_ids ? OPT	string[]
target_category_ids ? OPT	string[]
target_branch_ids ? OPT	string[]
excluded_product_ids ? OPT	string[]
excluded_category_ids ? OPT	string[]
discount_type ? legacy NOTE "alias"	string
max_discount_value ? legacy NOTE "alias"	number
created_by FK ?	string
created_at	datetime
updated_at	datetime

CouponClaim	
_id PK	string
coupon_id FK	string
user_id FK	string
claimed_at	datetime
status	string
used_order_id FK ?	string
created_at	datetime
updated_at	datetime

CouponUsage	
_id PK	string
coupon_id FK	string
user_id FK	string
order_id FK	string
discount_amount	number
used_at	datetime

ReturnRequestItem	
_id PK	string
return_request_id FK	string
order_item_id FK ?	string
product_id FK ?	string
quantity	number
reason ?	string
condition ?	string
"Optional extracted entity; may remain embedded"	NOTE

Promotion	
_id PK	string
title	string
description ?	string
type IDX	string
voucher_type ?	string
status IDX	string
start_date	datetime
end_date	datetime
is_active DEF true	boolean
priority DEF 0	number
scope ? NOTE "duplicated defs in schema"	string
target_product_ids ? NOTE "duplicated defs; targeting"	string[]
target_category_ids ? NOTE "duplicated defs; targeting"	string[]
target_branch_ids ? NOTE "duplicated defs; targeting"	string[]
excluded_product_ids ? OPT	string[]
excluded_category_ids ? OPT	string[]
is_auto_generated DEF false	boolean
source ? OPT	string
suggested_by_system DEF false	boolean
total_quantity ? OPT	number
remaining_quantity ? OPT	number
claimed_count DEF 0	number
hide_after_expired_hours ? OPT	number
auto_hide_after_expired DEF false	boolean
notification_sent DEF false	boolean
usage_limit ? OPT	number
max_redemptions ? OPT	number
usage_count DEF 0	number
usage_per_user ? OPT	number
min_order_amount ? OPT	number
min_quantity ? OPT	number
gift_quantity ? OPT	number
discount_value ? NOTE "alias mismatch: value"	number
value ? legacy NOTE "alias of discount_value"	number
max_discount_amount ? OPT	number
gift_product_id FK ? OPT	string
points_multiplier ? OPT	number
badge_text ? OPT	string
banner_image ? OPT	string
image ? OPT	string
banner_url ? OPT	string
claim_campaign ? OPT	string
stackable DEF false	boolean
created_by FK ?	string
created_at	datetime
updated_at	datetime

PromotionClaim	
_id PK	string
promotion_id FK	string
user_id FK	string
branch_id FK ?	string
claimed_at	datetime
status IDX	string
used_order_id FK ?	string
created_at	datetime
updated_at	datetime

PromotionUsage	
_id PK	string
promotion_id FK	string
user_id FK ?	string
order_id FK	string
discount_amount	number
created_at	datetime